



Investment Society
Imperial College London



Lecture 2

Valuation Methods and Metrics

Accounting & Valuation Workshop

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Slide deck inspired by Dr. Euan Doidge and Prof. Anthony Barrett (Department of Chemistry, Imperial College London)
Slide designs by Ahmad Danesh (Head of Editorial, Imperial College Investment Society 2021/22)

Background

Some context to introduce you to a concept

Model Example

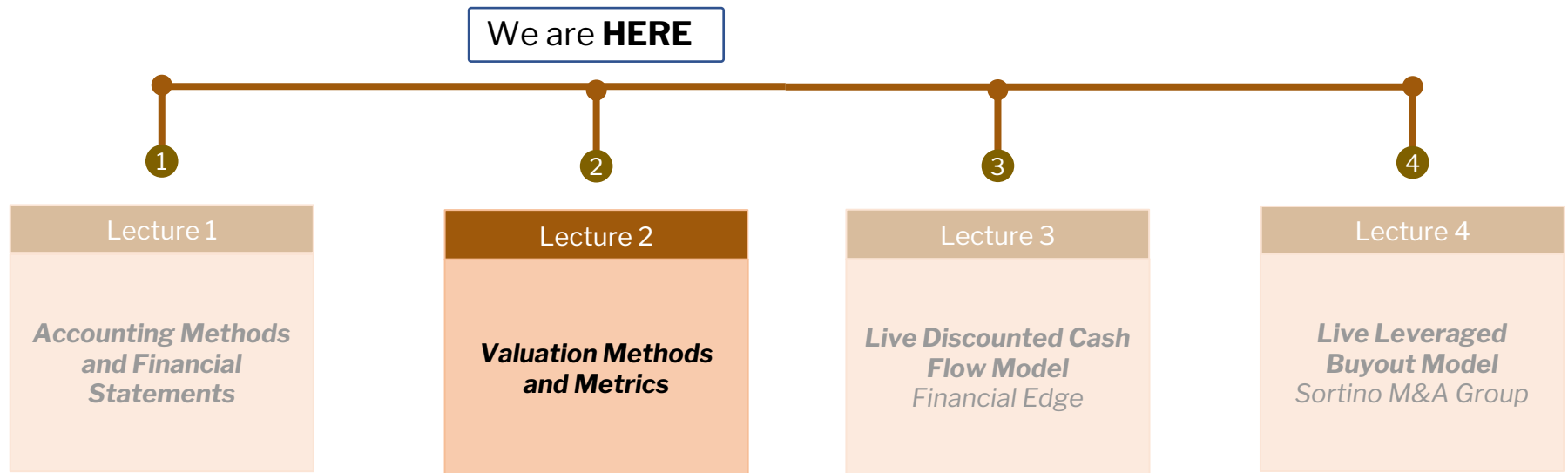
A fully detailed breakdown of an example

Taking it further

Real world application of a key concept covered

The slides are colour-coded to aid your understanding of the course and ensure that you are familiar with the demands of the content provided in each slide.

Lecture Timeline



Contents

What are we covering today?

Recap of Accounting
Introduction to Valuation
Precedent Transaction Analysis
Comparable Company Analysis
Discounted Cash Flow
Dividend Discount Model

Slides have been sent out via mailing lists.
Please let us know if you need any help.

Cash vs Accrual

Last week, you looked at different forms of accounting.

☞ Cash-based (tracking cash) and accrual-based (tracking activities) have their differences...

Cash-based

Used to track movements of cash.

Cannot go into debt.

Not a good indicator of how productive a business is.

Might be slower to grow the business.

Accrual-based

Used to monitor day-to-day business activities.

Good indicator of a company's actions.

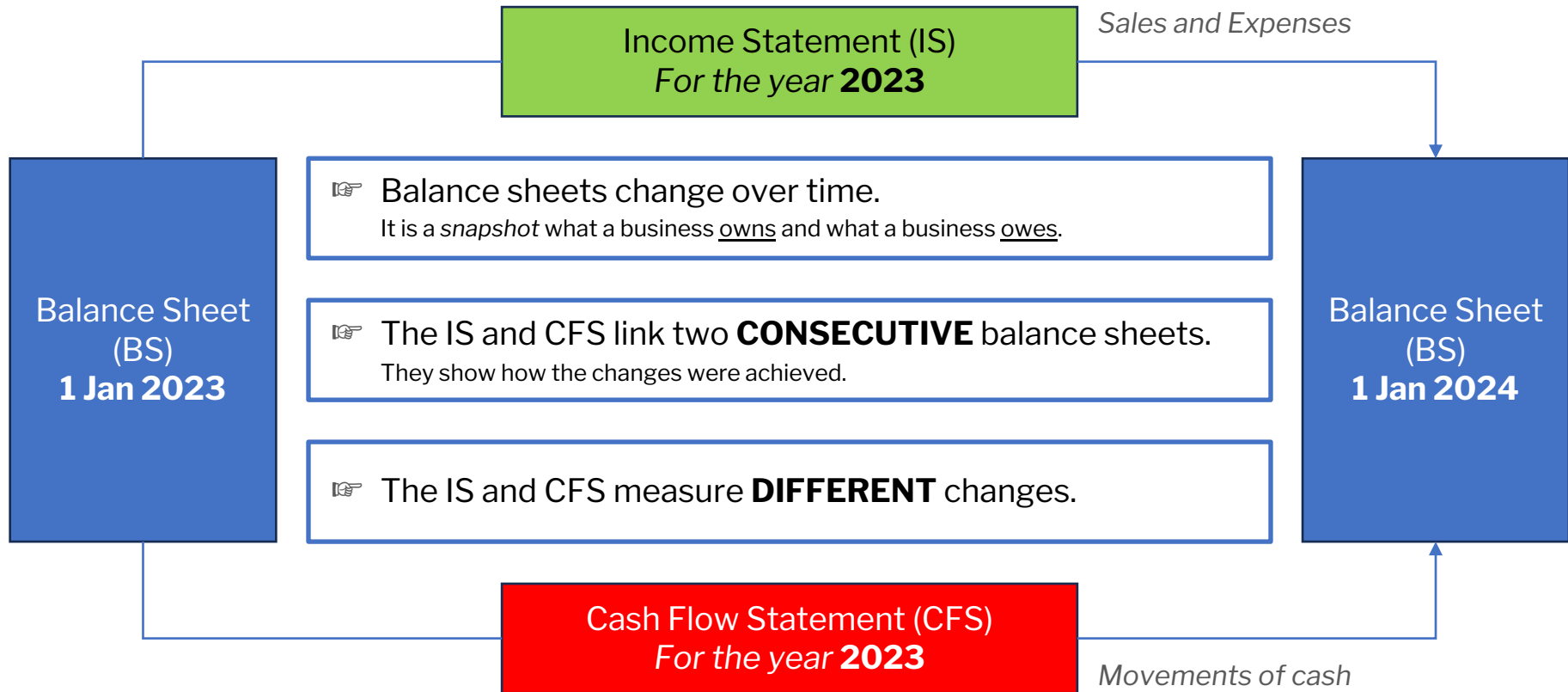
Adds potential complexity to recording transactions.

Allows businesses to buy items now and pay later.
i.e. act "on credit"

Only small-scale businesses are allowed to use cash-based accounting.

The three financial statements

What are they? How do they link together?



Income Statements look at sales and expenses.

Cash flow statements look at the movement of cash.

Balance Sheet

What is on the balance sheet?

👉 The balance sheet is a **SNAPSHOT** of the business at a moment in time.
It shows what the business owns and what the business owes.

👉 All balance sheets *balance* due to the fundamental equation of accounting...

$$\text{Total Assets} = \text{Total Liabilities} + \text{Shareholders' Equity}$$

👉 An asset is anything that can provide us some useful benefit.

👉 Liabilities are debts a company owes to third parties.
This could be banks, private investors etc.

👉 Shareholders' Equity is the value of the business that shareholders are entitled to when a business is closed down.
It is **NOT** simply the value of shares in the company.

The closure of a business is known as liquidation. You may also hear it referred to as 'ceased trading'.

Balance Sheet

We will be using this balance sheet.

Balance Sheet		
QT Auto		All values in USD \$
2023		
Assets		
Current Assets		
Cash	54	
Accounts receivable	46	
Inventory	230	
Short-term investments	400	
Total current assets		730.00
Non-current Assets		
Long-term investments	200	
Property, plant, and equipment	50	
Intangible assets	50	
Total non-current assets		300.00
Other Assets		
Deferred income tax	340	
Other	-	
Total Other Assets		340.00
Total Assets		1370.00
Liabilities		
Current Liabilities		
Accounts payable	30	
Short-term loans	40	
Income taxes payable	20	
Salaries and wages	30	
Common stock dividend payable	30	
Total current liabilities		150.00
Non-current Liabilities		
Long-term debt	150	
Other	20	
Total non-current liabilities		170.00
Total Liabilities		320.00
Shareholders' Equity		
Common Stock: 100 shares outstanding	1000	
Retained earnings	30	
Preferred Stock: 20 shares outstanding	20	
Total shareholder's equity		1050.00
Total Shareholders' Equity		1050.00

👉 **ALL** balance sheets are structured in the same order.

👉 Assets are listed in order of liquidity, most liquid first.

👉 Liquidity refers to how easy it is to sell something.
Items in a company's inventory (e.g. goods in stock) are easier to sell than e.g. property.

👉 For both assets and liabilities, *current* line items are listed **FIRST**.
These items can be transformed within 12 months.

👉 Entries in the assets and liabilities' section can be categorised in two ways...

Operational

Financial

This will be critical for later when calculating operating working capital.

Balance Sheet

We will be using this balance sheet.

Balance Sheet		
QT Auto		
2023		All values in USD \$
Assets		
Current Assets		
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Deferred income tax	340	
Other	-	
<i>Total Other Assets</i>		340.00
Total Assets		1370.00

Liabilities		
Current Liabilities		
Accounts payable	30	
Short-term loans	40	
Income taxes payable	20	
Salaries and wages	30	
Common stock dividend payable	30	
<i>Total current liabilities</i>		150.00
Non-current Liabilities		
Long-term debt	150	
Other	20	
<i>Total non-current liabilities</i>		170.00
Total Liabilities		320.00
Shareholders' Equity		
Common Stock: 100 shares outstanding	1000	
Retained earnings	30	
Preferred Stock: 20 shares outstanding	20	
<i>Total shareholder's equity</i>		1050.00
Total Shareholders' Equity		1050.00

Income Statement

What's recorded on the income statement?

☞ The income statement summarises the business's activities over a certain time window.
It works inline with the **accrual** method of accounting.

$$\text{Net Income} = \text{Total Revenue} - \text{Total Expenses}$$

☞ Total Revenue can be split into three parts...

Operating

Non-operating

Gains

☞ Total Expenses can also be expressed into three unique parts...

Operating

Non-operating

Losses

Income Statement

We will be using this income statement.

Income statement		
QT Auto		
2023		values in \$USD
Revenue		
Operating Revenue		
Gross sales	360.00	
Total operating revenue		360.00
Non-operating Revenue		
Interest	15.00	
Rent	25.00	
Total non-operating revenue		40.00
Total Revenue		400.00
Expenses		
Operating Expenses		
Cost of goods	50.00	
Employee Salary	40.00	
Sales returns and allowances	15.00	
Advertising	8.00	
Other	-	
Total operating expenses		113.00
Non-operating Expenses		
Amortisation	5.00	
Bank charges	5.00	
Charitable contributions	1.00	
Commissions	5.00	
Depreciation	4.00	
Insurance	7.00	
Interest	3.00	
Total non-operating expenses		30.00
Total Expenses		143.00
Other Income		
Gains on sale of assets	30.50	
Loss of changing location	100.00	
Total other income		-69.50
Pre-tax income		187.5
Income tax expense		37.5
Net Income		150

👉 **MOST** income statements are structured in the same order.

👉 Sales from selling business's goods are given first.
These are known as '*operating revenues*' as they are the main items the business sells.

👉 Non-operating revenue is given lower priority e.g. interest or rent income.
These are not (generally) part of the business's main operational goals.

👉 You get further away from the core product as you go down the statement.

👉 Tax is calculated last, after all other expenses are accounted for.
A tax rate is required, normally given in %.

The income statement links two balance sheets together!
You always begin your analysis with the income statement **FIRST**.

This will be critical for later when calculating operating working capital.

Cash Flow Statement

What does a cash flow statement look like again?

☞ Cash flow statements (**CFS**) tell me how the amount of cash in a business has changed in the given time period.

☞ Cash flow statements work on the cash-based accounting process.
It allows tracking of money in the business's bank account whilst accrual accounting is used in the IS and BS.

☞ We take last year's figure as our **B**eginning figure, **A**dd or **S**ubtract to it and arrive at an **E**nding value for this year.
This is known as **BASE** analysis.

☞ Cash flow statements can be broken down into three separate categories...

Operating

Cash related to the main business.

Investing

Cash spent or made on investments.

Financing

Cash used for financing business operations.

Cash Flow Statement

We will be using this cash flow statement.

Cash Flow Statement	
QT Auto	
2023	All values in USD \$
Operating Cash Flow	
Net Earnings	150.00
Depreciation & Amortisation	2.00
Net Cash from Operations	
	152.00
Investing Cash Flow	
Income from Property Investments	10.00
Purchase of ABC shares	-3.00
Net Cash from Investing	
	7.00

Financing Cash Flow	
Repayment of debt	20.00
Repayment of equity	148.00
Net Cash from Financing	
	-168.00
Net change in Cash	
	-9.00
Opening Cash Balance	200.00
Closing Cash Balance	
	191.00

Let's begin...



Dev Shah

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Lecture 2

Valuation Methods and Metrics

Valuation

Why do we need to value companies?

👉 Investors want to buy a company based on its **FUTURE** potential.

👉 They will estimate how the three financial statements will appear in the foreseeable future.
This is known as *projection* or *(financial) forecasting*.

👉 Analysts will then use various strategies to arrive at a price point.

👉 An analyst looks at various features of the business in question, including but not limited to...

Management

Capital Structure

Debt or Equity

Future potential

(High profit-making possibilities)

👉 This is critical research that is required to ensure investors can make the best possible decision.
This is known as performing *due-diligence*.

Think about the due-diligence you did when selecting your University of choice.

Types of Valuation

There are two main ideas.

Absolute Valuation

Aims to find the **intrinsic** or **true** value of an investment

Based on fundamentals of a **single** company

Not worried about the performance of other companies

Uses factors e.g. growth rate, cash flow, dividends etc.

Models such as DCF and DDM fall into this category

Relative Valuation

Compare the company of interest to **other** companies

(Typically) of a similar size and in the same industry

Involves calculating multiples and ratios

Use ratios as basis of comparison

e.g. P/E, EV/EBITDA, debt-to-equity

We will discuss these concepts shortly.

Absolute and Relative Valuation

Let's think of a simple example to show this.

👉 Say I want to purchase QT Chemicals. I would first look at its profits for the last year.
Let's consider QT Chemicals' profits to be £100M each year for the past 5 years.

👉 I will have to pay much more than £100M to buy QT Chemicals since that is the *annual* profit.

👉 I will need to pay a figure that is a multiple of £100M to convince the owners to sell e.g. on 4.4x at £440M.
This is known as paying on a profit **MULTIPLE**.

👉 To obtain this value of £440M, I would need to do some detailed predictions of its operations in the future.
This is what is meant by *absolute valuation*.

👉 I could also compare QT Chemicals with e.g. UCL Chemicals for comparison.
If UCL Chemicals is valued for £400M, this can be used as a ballpark figure.

👉 Comparison of companies in the same industry, geography, size etc. is a valid approach.
This is what is meant by *relative valuation* i.e. *relative* to another company.

Valuation Methods

What are some valuation method used?

Precedent Transaction Analysis

Comparable Company Analysis

Discounted Cash Flow

Gordon Growth Model

Net Asset Value

We normally use all these methods and compare the results.

👉 Remember that valuation methods are an **ESTIMATION**, not an exact value.

👉 Your valuation results are dependent on the quality of your assumptions.

👉 Valuation models become less reliable as you extend the timespan of the predictions.
It is unlikely that a forecast 5+ years in the future will hold due to macroeconomic changes etc.

Precedent Transaction Analysis

What is precedent transaction analysis?

M&A = Mergers and Acquisitions

 **Precedent Transaction Analysis (PTA)** - A relative method of valuing a company by looking at similar companies bought and sold through past M&A deals.

Looks at price paid for similar companies in the past

Uses publicly available information
to create estimates and multiples

Acts as indicator of company value
(if it were to be acquired)

Need to understand relevance of transactions

 This tool isn't foolproof as no two companies are exactly ever the same.

 We use PTA in combination with other methods to check whether our value makes sense.
i.e. in the same ballpark range.

It's straightforward and easy! It frequently happens in M&A deals with football clubs.

Precedent Transaction Analysis

M&A = Mergers and Acquisitions

 **Precedent Transaction Analysis (PTA)** - A relative method of valuing a company by look at similar companies bought and sold through past M&A deals.

Limitations

PTA can be very difficult to conduct

The market at the time of the deal will differ to today
e.g. market condition, size, industry, economic climate etc.

Lack of quality information available.

Difficult to make required adjustments.

Benefits

Can be used as a benchmark price

Information is easily accessible and uncomplicated

Comparable Company Analysis

What is Comparable Company Analysis?

☞ **Comparable Company Analysis (CCA)** - A relative valuation method where we look at another similar company to calculate a fair value for our own company.

☞ We look at e.g. the value of a similar company as a multiple of its earnings.

☞ We assume this multiple is the same for our company as well.

A CCA is very easy and straightforward to implement.

☞ We start by establishing a group of similar companies consisting of similar size, industry, and/or region.

To do a CCA, we also need to first look at certain statistics and metrics and see how they behave.

Market Capitalisation

What is market capitalization?

👉 **Market Capitalisation ('market cap')** - Total market value of a company's shares.

Market Cap = Share price x Number of shares outstanding

👉 Market capitalization looks at how much the stock market currently values the equity in a company.

👉 Market cap is also known as the market value of a company.

👉 Very useful for finding the market dominance of certain companies in their respective industries.
It is very useful as a quick comparison between companies in the same sector.

Market Capitalisation

Going back to our truck manufacturer...

☞ The current share price of a company can be found on the stock exchange that it is currently listed on.

☞ The number of shares issued by the company is found on the balance sheet in the shareholders' equity section.

Shareholders' Equity		
Common Stock: 100 shares outstanding	100	
Retained earnings	30	
Preferred Stock: 20 shares outstanding	20	
<i>Total shareholder's equity</i>		1050.00

Suppose QT Auto's common shares are currently trading at £20 on the London Stock Exchange. As seen on their balance sheet, they have issued 120 shares, hence their current market cap is $120 \times 20 = £2,400$.

We only consider 'preferred shares' if they are **PUBLICLY** listed!

Enterprise Value

A lot of ways to calculate value.

👉 **Enterprise Value (EV)** – An alternative measure of a company's total value which incorporates debt, preferred equity, minority interest and cash or cash equivalents.

$$\text{EV} = \text{Market Cap} + \text{Total Debt} + \text{Minority Interest} + \text{Preferred Equity} - \text{Cash or Cash Equivalents}$$

Total Debt – Sum of long-term and short-term debt

Minority Interest – Stake less than 50% in a subsidiary which is controlled by parent company

Preferred Equity – Equity which has a higher priority of a company's cash flow

Cash or Cash Equivalents – Total amount of cash and investments that can be readily converted into cash

👉 This is because in order to purchase a company you would have to pay the **market price** for its shares, but also have enough cash to pay off its **debt**.

👉 Since cash offers **no value** to the business and can be used to pay off existing debt, it is subtracted to calculate the EV.

EBIT

This is a bit more nuanced.

👉 **EBIT** – Earnings before interest and tax used as an indicator of profitability

$$\text{EBIT} = \text{Revenue} - \text{Expenses} + \text{Interest} + \text{Taxes}$$

Focuses solely on company's ability to produce operating profit

Ignores variables such as tax burden and capital structure

Used in lots in comp analysis

Comes with its limitations (Depreciation and Amortization)

If a tax break is received by a company with large amounts of debt it is a very useful metric

EBIT

Lets' look at an example.

👉 **EBIT** – Earnings before interest and tax used as an indicator of profitability

$$\text{EBIT} = \text{Revenue} - \text{Expenses} + \text{Interest} + \text{Taxes}$$

Repayment of debt (interest)	20.00
------------------------------	-------

Income Tax Expense	37.50
--------------------	-------

We can find the interest and tax payments on the income statement and cash flow statement and recall the net income is £150. So we see the EBIT for QT Auto is $\text{£150} + \text{£20} + \text{£37.50} = \text{\$207.50}$

👉 EBIT ignores costs not directly associated with the company's operations and hence is sometimes seen as a more accurate measurement of earnings.

EBITDA

Even more nuanced.

👉 **EBITDA**– Earnings before interest, tax, depreciation, and amortization used as an indicator of profitability

$$\text{EBITDA} = \text{EBIT} + \text{Depreciation} + \text{Amortisation}$$

👉 **Remember:** Depreciation and amortization refer to the loss of value of assets over time.

👉 This cost is seen as something that is out of the control of the business and hence is excluded when calculating earnings, as they don't contribute to the main business operations.

Useful when looking at companies with large amounts of fixed assets

E.g. automotive companies

EBITDA

Let's look at an example.

👉 **EBITDA**– Earnings before interest, tax, depreciation, and amortization used as an indicator of profitability

$$\text{EBITDA} = \text{EBIT} + \text{Depreciation} + \text{Amortisation}$$

Depreciation & Amortisation	2.00
-----------------------------	------

We recall the EBIT of QT Auto as £207.50. Depreciation and amortisation can be found on the income statement as £2. Hence we add this value to obtain an EBITDA of £209.50

👉 Note, lots of investors dislike the use of EBITDA as they see it as a way of falsely inflating a company's earnings.

Warren Buffett notoriously hates the use of EBITDA.

Valuation Metrics

How do we measure performance?

☞ There are many common metrics used to measure the performance of a company.

☞ Most of these metrics measure the **profitability** of a company, allowing us to see how much it costs for an investor to invest in it.

☞ There are many ways to measure this and hence many metrics people use, such as:

P/E ratio

EV / EBITDA

Return on equity

☞ They are great for comparing companies in similar **sectors** or during similar periods of **growth**.

Earnings per Share

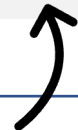
One of the most fundamental tools.

👉 **Earnings per Share (EPS)** – indicates how much money a company makes for each share of its stock.

Earnings per share = (Net Income – Preferred Dividends) / Avg Number of Common Shares Out

One of the most important metrics for profitability

Accurate measure for shares outstanding.



Not the most useful on its own but can be combined for more useful metrics.

👉 We subtract **preferred dividends** as common stockholders don't have access to these earnings.

Earnings per Share

One of the most fundamental tools.

👉 **Earnings per Share (EPS)** – indicates how much money a company makes for each share of its stock

Earnings per share = (Net Income – Preferred Dividends) / Average Number of Common Shares Out

Net Income	150
-------------------	-----

Common Stock: 100 shares outstanding	1000
Retained earnings	30
Preferred Stock: 20 shares outstanding	20

For QT Auto we see they have a net income of £150 and 100 shares outstanding. Therefore their current earnings per share is $\text{£}150/100 = \text{£}1.50$.

👉 Using EPS is a quick way of assessing the financial prospects of a company.

👉 Hence, some companies buy back shares to improve this metric without increasing their income. With the hope of attracting more investors.

P/E Ratio

You're going to see this one a lot...

👉 **Price-to-Earnings Ratio (P/E)** – The price people are willing to pay for each unit of profit a company generates.

$$\text{P/E ratio} = \text{Share price} / \text{Earnings per share}$$

Benefits

Can help determine if a company is over or under valued

Used to compare similar industries

Limitations

Valuations and growth of companies varies wildly between sectors and companies

Needs to be used with other metrics

P/E Ratio

What does an example look like?

- 👉 In order to calculate the P/E ratio you require the share price from the stock exchange, the net income (earnings) from the income statement and the number of shares outstanding from the balance sheet.

We know from the previous calculation that QT Auto have an earnings per share of £1.50. From the stock exchange we see their shares are currently trading at £20, therefore QT Auto have a P/E ratio of $\text{£20} / \text{£1.50} = 13.3$.

- 👉 One issue with a P/E ratio is they cannot be used for companies that are not currently generating profits. For this reason we use something slightly different.

P/S Ratio

What is price to sales?

👉 **Price-to-Sales Ratio (P/S)** – The price people are willing to pay for each unit of revenue (sales) a company generates

$$\text{P/S ratio} = \text{Share price} / \text{Revenue per share}$$

Benefits

Allows us to compare companies which are not profitable yet

Can be a good indicator of growth

Limitations

Doesn't take into account costs

Doesn't look at individual components of revenue

P/S Ratio

Let's look at an example.

- ☞ Same procedure as P/E ratio, except we only need to look at the revenue section of the income statement.

Total Revenue	400.00
---------------	--------

Looking at the income statement again QT Auto generated £400 in revenue. We know already from the balance sheet they have issued 100 shares, therefore they have a revenue per share of $\text{£}400/100 = \text{£}4$. Their share price stays the same at £20 and so QT Coffee have a P/S ratio of $\text{£}20 / \text{£}4 = 5$.

- ☞ Note that P/S ratio should **always be less** P/E ratio, since revenues are always greater than profits.

- ☞ Although the P/S ratio doesn't take costs into account, this type of analysis might be ok for some companies but not for a majority of them.

This is typically fine for companies just entering the market.

EV/ EBITDA

How do we use these in reality?

👉 We can define a similar ratio to the P/E ratio, but altering each part slightly.

👉 Instead of share price we use **EV** and instead of earnings we substitute for **EBITDA**.

👉 This is because:

$$\frac{EV}{EBITDA} = \frac{\text{Market Cap} + \text{Total Debt} - \text{Cash or cash equivalents}}{\text{Earnings} + \text{Interest} + \text{Taxes} + D \& A},$$

$$PE \text{ ratio} = \frac{\text{Market Cap}}{\text{Earnings}}$$

👉 EV/EBITDA is sometimes seen as more **accurate** than P/E.

👉 This is because EV is seen as a better way of determining equity value rather than market cap, and EBITDA a more meaningful way to measure earnings.

Debt to Equity Ratio

Pretty straightforward one here...

👉 **Debt-to-Equity** ratio just looks at how much debt a company has compared to the value of its equity.

$$\text{Debt-to-Equity ratio} = \text{Total Liabilities} / \text{Shareholders' Equity}$$

👉 This ratio compares how much money a company is borrowing to the value of its equity. Essentially seeing if the value of equity can cover the cost of debt.

👉 If this value is too high this could be risky for shareholders, as more cash will go to paying interest payments on debt, rather than dividends.

Debt to Equity Ratio

What does the calculation look like?

👉 For this ratio we just need the **Liabilities** and **Shareholders' Equity** sections of the balance sheet.

Total Liabilities	320.00
-------------------	--------

Total Shareholders' Equity	1050.00
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Looking at the balance sheet we see that QT Auto has total liabilities worth £320. We also see that their shareholders' equity is worth £1050. Therefore their debt-to-equity ratio is $\text{£}320 / \text{£}1050 = 0.30$.

👉 This is a great metric for comparing the **leverage** used by different companies in order to sustain their business.

*Leverage refers to the **borrowing** of funds in order to do financial transactions.*

👉 It can also help determine lower-risk investments, as a high debt-to-equity ratio could suggest that a company would have issues **incurring more debt** and **maintaining payments** to creditors.

Return on Equity

How do I measure my expected returns?

👉 **Return on Equity** is another metric used to measure a company's profitability.

👉 It is calculated in the following way:

$$\text{Return on Equity} = \text{Net Income} / \text{Shareholders' Equity}$$

👉 Sometimes net income is used **before** the payment of dividends, as this would represent returns to investors.

👉 This measures how efficiently a firm's management uses money from their shareholders to **generate profit**.

Return on Equity

How can we calculate this?

👉 In this case we just need the **income statement** and the **balance sheet**.

Net Income	150	Total Shareholders' Equity	1050.00
Repayment of equity (dividends)		148.00	

We see that QT Auto had a net income of £150 and paid £148 in dividends. If we add this back we get an income of £298. Shareholders' equity is worth £1050 and so the return on equity in this period is $\text{£298} / \text{£1050} = 28\%$

👉 A good return on equity is seen as one that increases **steadily over time**, instilling confidence in shareholders that their money is being used wisely.

Let's continue...



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Lecture 2

Valuation Methods and Metrics

Discounted Cash Flow

Now, how do we find the fair price of a company?

👉 **Discounted Cash Flow** (DCF) is a very common method for finding the current value of a company.

👉 A discounted cash flow model values a company based on projected future cash flows.

👉 You **discount** future cash flows based on different rates such as WACC or APV, for example.

👉 The theory behind a DCF is as a company grows and becomes more profitable, it should generate greater cash flows.

👉 Therefore this model works out how much we should pay for those cash **future** flows.

*As well as beyond the time period of the **DCF**!*

Free Cash Flow

We first need to look at free cash flow...

☞ There are two types of **free cash flow** (FCF):

Unlevered FCF

Free cash flow to the **firm** (FCFF)

Cash flow before any interest payments on debt

Levered FCF

Free cash flow to **equity** (FCFE)

Cash flow after company meets its debt obligations

☞ For a DCF we use **unlevered** free cash flow as it can be difficult to predict a company's future debt

☞ There are many different ways to measure free cash flow, especially depending on what data you have available.

Free Cash Flow

How can cash be free?

- 👉 Free cash flow refers to the amount cash left over from operating cash flows after capital expenditure has been paid for.

$$\text{FCF} = \text{Operating Cash Flow} - \text{Capital Expenditure}$$

- 👉 Operating cash flow is found on the **cash flow statement** and capital expenditure is found in cash flow from **investing activities** on the same statement.

- 👉 **Capital expenditure** refers to the amount of funds spent to maintain a fixed asset, such as property or land.

- 👉 We can derive free cash flow by just using the balance sheet and income statement:

$$\text{FCF} = \text{Net Income} + \text{Non-Cash Expenses} - \text{Change in NWC} - \text{Capital Expenditures}$$

NWC (Net Working Capital) = Current Assets – Current Liabilities

Time Value of Money

But why do we need to discount cash flows?

☞ This is because money **loses value** over time.

If I gave you the option of £100 now or £100 in a year, almost everyone would pick the £100 now, and this is for two main reasons:

Inflation

Opportunity Cost

☞ **Opportunity cost** refers to cost of not being able to choose the next best alternative.

☞ We want to **discount** the future value of cash flows according to these costs in order to find their present values.

Time Value of Money

How does it work in practice?

Suppose in a year's time you want to have £100 in your bank account. The interest rate offered to you by your bank is 5%. How do you work out what you need to put in your bank now?

👉 Well you need to discount the future value to the present:

Today ●—————● In 1 year

You need £95.23 to get £100 in a year at 5% interest.

$$\frac{£100}{1.05}$$

You receive £100.

👉 Therefore the present value of £100 in a year is £95.23 at 5% annual interest.

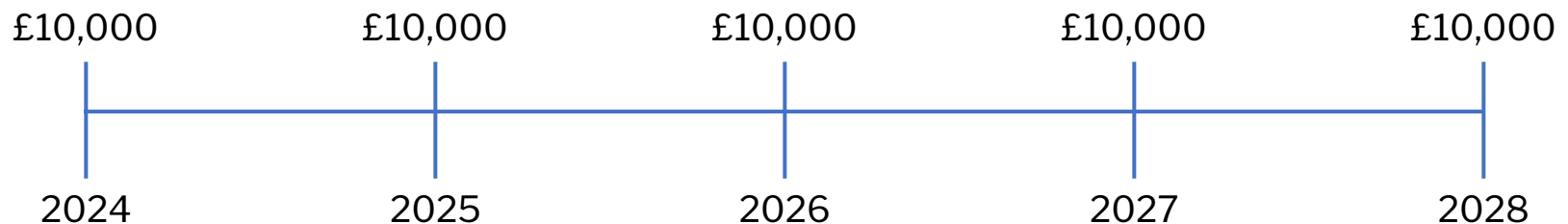
Discounted Cash Flow

What does the DCF model tell us?

👉 So we discounted our future cash flow by our rate of return, the interest rate, to find the present value.

👉 Now we extend this idea for multiple cash flows:

Suppose QT Auto is projected to earn £10k in net profit every year for the next five years. How much should you pay now for those cash flows for a 10% annual return?



👉 We need to find the present value of each year of cash flow.

Discounted Cash Flow

What does the model tell us?

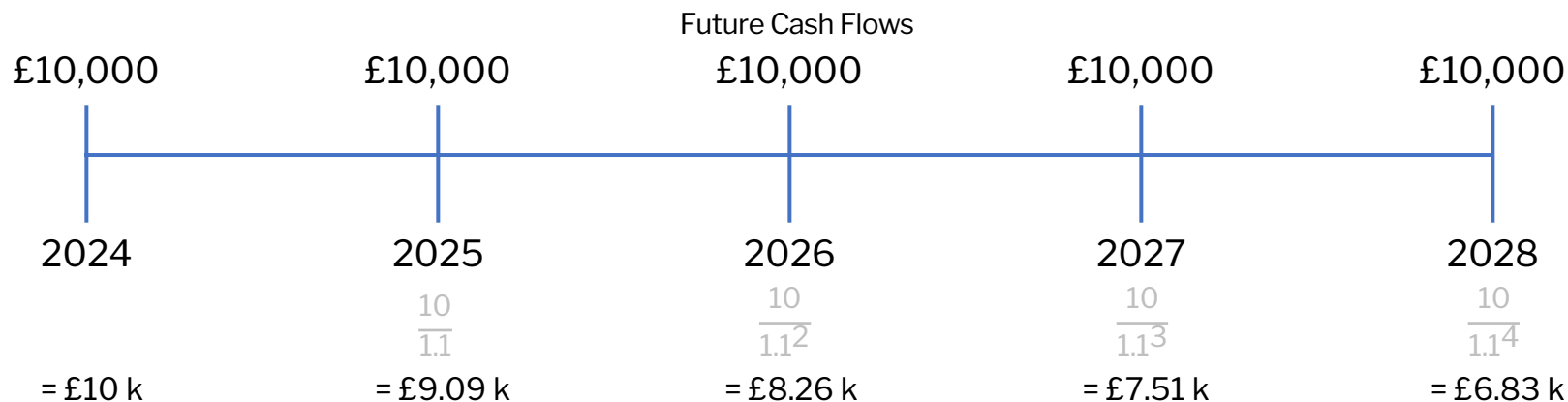
👉 The 2024 cash flow should stay the same since we are in this year, so present value is £10,000.

👉 For the 2025 cash flow we should discount it by 10%, so we get

$$\text{Present value} = £10\text{k} \div 1.1 = \textbf{£9.09 k}$$

👉 For the 2026 cash flow we should discount it by 10% twice since it is 2 years away, so we get

$$\text{Present value} = £10\text{k} \div 1.1^2 = \textbf{£8.26 k}$$



Present Value (of Future Cash Flows)

Discounted Cash Flow

How do we work out present values?

☞ The total present value for all years of cash flow is:

$$\begin{aligned}\text{Sum of present values} &= \text{£10 k} & + \text{£10k}/1.1 & + \text{£10k}/(1.1^2) & + \text{£10k}/(1.1^3) & + \text{£10k}/(1.1^4) \\ &= \text{£10 k} & + \text{£9.09 k} & + \text{£8.26 k} & + \text{£7.51 k} & + \text{£6.83 k} \\ &= \text{£41.7 k}\end{aligned}$$

☞ So for £50k of future cash flows we should pay £41.7 k.

$$\text{Present Value} = \frac{\text{Cash Flow 1}}{(1+r)} + \frac{\text{Cash Flow 2}}{(1+r)^2} + \dots + \frac{\text{Cash Flow } n}{(1+r)^n} \text{ for discounting } n \text{ years of cash flow at a rate of } r.$$

☞ Some of you may have noticed this sum is equivalent to a geometric series.

$$S_n = \frac{a(1-r^n)}{(1-r)},$$

where n = number of terms, a = first term and r = common ratio

$$\text{Present Value} = \frac{10 \left(1 - \frac{1}{1.1^5} \right)}{1 - \frac{1}{1.1}} = \text{£41.7 k}$$

WACC

WACC is that?

☞ For the **discount rate** we used our expected return, but there is a different discount rate more commonly used.

☞ **WACC** refers to the weighted average cost of capital.

☞ Remember from the income statement, there are two ways for a company to generate capital:

Take on debt

Sell some equity

☞ So the WACC calculated how much it costs the company to generate this capital, and then weighs it accordingly.

For example, if QT Coffee's cost of capital is 20% equity and 80% debt, then the WACC is $\frac{(20 \times C_E) + (80 \times C_D)}{100}$, where C_E refers to the cost of equity and C_D is the cost of debt. Hence $C_E + C_D$ is the total cost of capital.

WACC

How do we actually derive it?

☞ In particular, we can formulate these proportions of equity and debt.

$$\left(\frac{E}{E+D} \times C_E\right) + \left(\frac{D}{E+D} \times C_D \times [1 - T]\right) = WACC$$

☞ Where **E** refers to the total value of the company's equity, and **D** the value of its debt.

☞ Let **V** = E + D

$$\left(\frac{E}{V} \times C_E\right) + \left(\frac{D}{V} \times C_D \times [1 - T]\right) = WACC$$

☞ The **cost of equity** (C_E) refers to the expected rate of return from shareholders, whilst the **cost of debt** (C_D) is the interest rate on the principal amount of debt.

WACC

How do we actually derive it?

☞ Debt is sometimes more useful than equity as it provides what is known as **tax-shielding**.

☞ This is because interest is a **tax-deductible expense**, meaning it is taken into account before the tax due to be paid is calculated.

Interest	10,000
Net Income before tax	100,000
Taxes (20%)	20,000
Net income	80,000

☞ If this company didn't have any interest payments, they would pay \$22,000 in taxes.
Their net income would be \$110,000

☞ The \$10,000 interest payment saves \$2,000 in taxes.

☞ The cost of debt is the interest paid per year – any tax saved.

☞ The cost of debt is therefore $\$10k (1 - 0.2) = \8000 .

20% tax rate



WACC

What does the equation look like?

☞ Now we have derived the final form of the weighted-average cost-of-capital formula.

$$\left(\frac{E}{E + D} \times C_E \right) + \left(\frac{D}{E + D} \right) (C_D \times [1 - T]) = WACC$$

☞ Where **T** is the tax rate.

☞ The WACC weights the proportions of equity and debt in a company to give a better representation of the cost to the business to **raise capital**.

☞ It is very commonly used as a **discount rate** in DCF models.

☞ This is because it is seen as a more accurate measure for the return an investor should expect.

Cost of Debt

How can we calculate the cost of debt?

👉 The **cost of debt** for a company is the rate a company is paying for its debt.

👉 Values for this can be found from the company's **income statement** and **balance sheet**.

$$\text{Cost of Debt} = \frac{\text{Interest expense}}{\text{Total debt}}$$

👉 The interest expense can be found from the income statement

👉 The total debt can be found from the liabilities section of the balance sheet

Cost of Equity

What model can we use to calculate the cost of equity?

☞ **The Capital Asset Pricing Model (CAPM)** works out the expected return of an investment..

☞ It utilizes past data and performance to extrapolate what might happen to the future value of an asset.

☞ First we define a variable called **beta**, which denotes the riskiness of our asset.

$\beta = 0$ Risk-free investment, behaves like government bond.

$\beta < 1$ Less risky than other typical market assets.

$\beta = 1$ Behaves like the average market asset, in terms of risk.

$\beta > 1$ More risky than the typical market asset.

☞ Risk is defined as the chance an investment's return will differ from the expected return.

☞ Therefore the higher the risk, the higher the return investors want.

Capital Asset Pricing Model

Risk-free? Sounds good to me.

☞ The CAPM then compares the return of this asset, to the return of a risk-free investment.

☞ We denote this as R_f .

R_f = return on risk-free investment

☞ Typically, a risk-free investment would be a **government bond**.

☞ Government bonds are where you lend out money to the government, in return for interest payments on that money.

☞ These are seen as risk-free as the risk of a government defaulting on your loan is incredibly low.

US 10 Year Treasury: 4.06%

UK 10 Year Gilt: 3.88%

Capital Asset Pricing Model

We need to look at some more variables.

☞ The next variable in this model is the market return, denoted R_m .

☞ This measures the return of the typical asset in the market you are trading.

R_m = Market return

☞ This could be stocks, fixed incomes, commodities, real estate, etc.

☞ ' $R_m - R_f$ ' is known as the '**market risk premium**'. It is essentially the extra expected return from this market compared to the risk-free investment as a result of the extra risk taken.

$R_m - R_f$ = Market risk premium

Capital Asset Pricing Model

How do I calculate my profit?

☞ The model then predicts the **expected return** on an investment.

☞ Essentially saying your expected return is equal to the return of the market plus the risk of the asset multiplied by the market risk premium.

☞ The model says there is a **linear relationship** between the expected return and the risk of an asset

$$R_a = R_f + \beta(R_m - R_f)$$

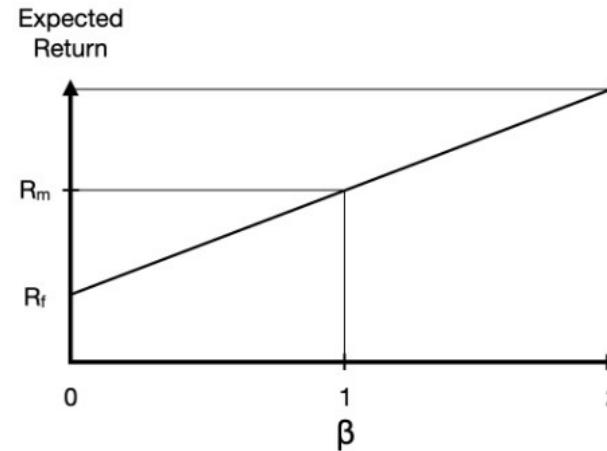
☞ Where **R_a** denotes the expected return of the asset.

Suppose we would like to invest in QT Auto. As QT Auto is listed on the LSE, we see the average annual return for stocks is about 7.5%. I can currently purchase an annual government bond with a yield of 4.5%. Due to the nature of the business of QT Auto I have assessed they have a beta value of 0.5. Hence my expected annual return $R_a = 4.5\% + 0.5 (7.5\% - 4.5\%) = 6\%$

Capital Asset Pricing Model

What are the issues with CAPM?

$$R_a = R_f + \beta(R_m - R_f)$$



👉 **CAPM** is not always reliable, since it extrapolates past data to predict future returns.

👉 As you become a seasoned investor you realise past performance is not always a good indicator of future performance.

👉 CAPM also assumes government bonds are risk free investments.

👉 In most cases the chances of a government defaulting on their loans is very small. However, there will continue to be instances where that happens.

Terminal Value

Not quite done with DCFs.

☞ We cannot predict future cash flows **forever**, so we define something called the terminal value.

☞ This is the value of the company beyond the period of time that we can predict the cash flows.

☞ A very easy way to calculate terminal values is to use an **exit multiple**.

$$\text{TV} = \text{Last period multiple} \times \text{Exit multiple}$$

☞ Last period multiple refers to some metric that measures the **profitability** of the company, typically EBITDA or EBIT.

☞ We then multiply this by some exit multiple to obtain the terminal value. Since we are using EBITDA, a good exit multiple would be EV/EBITDA.

Terminal Value

How do we calculate this?

$$T = \text{Last period EBITDA} \times \text{Average EV/EBITDA}$$

👉 We estimate the **last period EBITDA** using our projected cash flows for the company and any other assumptions.

👉 The **EV/EBITDA** is then obtained usually by looking at the current average for the sector that the company is in.

We calculated the EBITDA for QT Auto to be £309. According to our DCF cash flows are predicted to increase for QT Auto 5% annually. Hence, if we conduct our DCF for 3 years then after 3 years EBITDA = $£309 \times 1.05^3 = £357.71$. If the current average EV/EBITDA is 25, then the terminal value is $£357.71 \times 25 = £8942$.

$$\text{Company value} = \text{DCF}(n) + T$$

👉 Where n refers to the number of periods the DCF is calculated for.

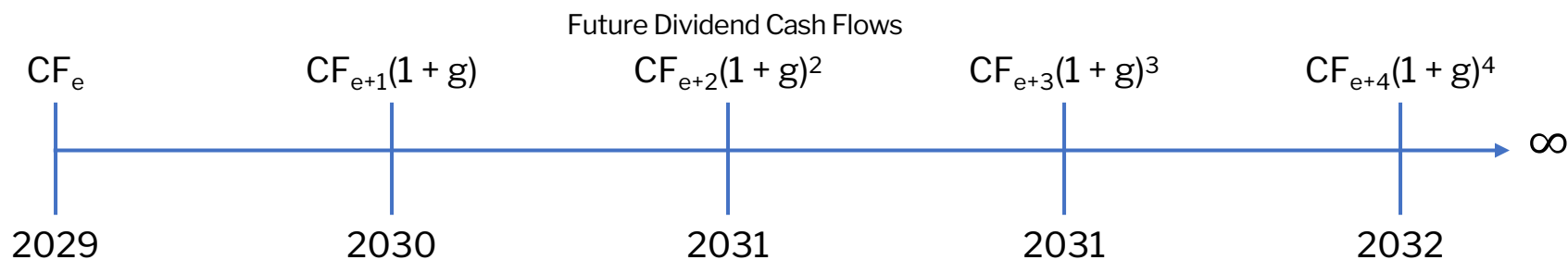
Gordon Growth Model

What is the Gordon Growth Model?

☞ This is another method which can be used to calculate the terminal value of a company

☞ We take a growth rate g which we believe the company can maintain indefinitely

☞ We assume that the company's cash flow will continue to grow at this rate beyond the end of the forecasting period

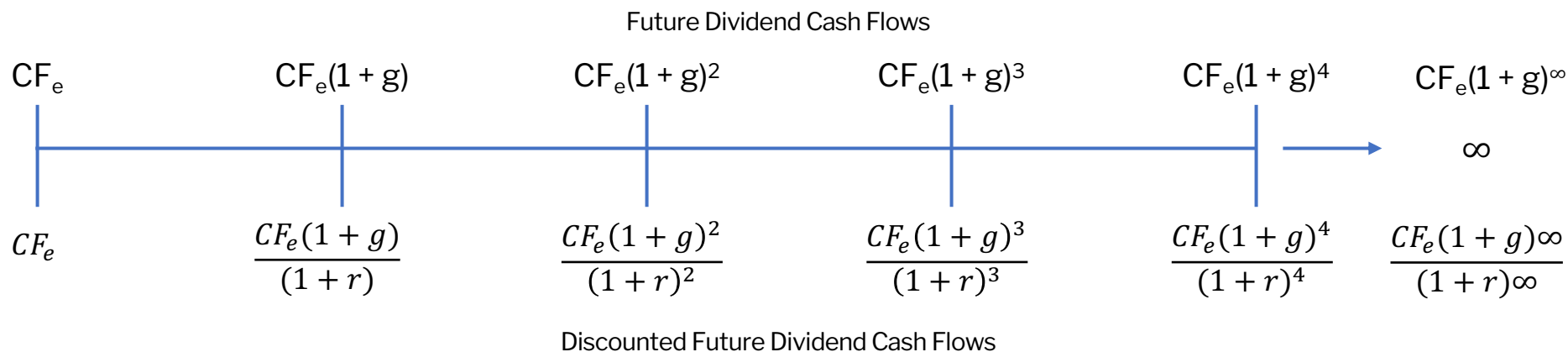


☞ We normally take g as 2% since it is less than the growth rate of the **global economy**.

Gordon Growth Model

How do we derive the value?

But we can then **discount** these future cash flows to get their present values:



We need our discount rate **r**, to be greater than the growth rate **g**. This is because otherwise the value of the company's dividends would be worth infinity.

Using the sum to infinity formula we can now workout the value of the company..

$$S_\infty = \frac{a}{1-r}, \text{ where, } a = \text{first term and } r = \text{common ratio, } |r| < 1$$

Gordon Growth Model

Ah, that doesn't look too bad.

☞ In our case:

$$a = \frac{CF_e(1+g)}{(1+r)}, \quad r = \frac{(1+g)}{(1+r)}$$

☞ And so:

$$S = \frac{\frac{CF_e(1+g)}{(1+r)}}{1 - \frac{(1+g)}{(1+r)}} = \frac{\frac{CF_e(1+g)}{(1+r)}}{\frac{(r-g)}{(1+r)}} = \frac{CF_e(1+g)}{(1+r)} \times \frac{(1+r)}{(r-g)} = \frac{CF_e(1+g)}{(r-g)}$$

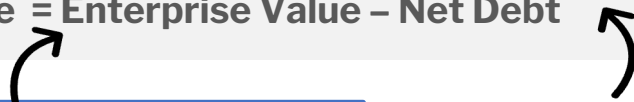
$$\mathbf{P} = \frac{CF_e(1+g)}{(r-g)}$$

- ☞ Where **P** is the present value of the company.
- ☞ **CF_e** is the cash flow at the end of the forecasting period.
- ☞ **r** is the discount rate.
- ☞ **g** is the growth rate.

Terminal Value

Lost? Let's take a step back

👉 We are trying to calculate the equity value of the company

$$\text{Equity Value} = \text{Enterprise Value} - \text{Net Debt}$$


👉 Enterprise value is the sum of the discounted cash flows and terminal value of the company

$$\text{Net Debt} = \text{Total Debt} - \text{Cash}$$

$$\text{Enterprise Value} = \frac{\text{FCF}_1}{(1+r)} + \frac{\text{FCF}_2}{(1+r)^2} + \dots + \frac{\text{FCF}_n}{(1+r)^n} + \frac{\text{TV}}{(1+r)^n}$$


👉 TV is calculate either using the **Exit Multiple** or **Gordon Growth Model**

Price per Share

How to we calculate the share price?

☞ To calculate the value per share we divide equity value by the number of shares outstanding

$$\text{Share Price} = \text{Equity Value} / \text{Number of Shares Outstanding}$$

☞ This will give you the **intrinsic price per share.**

☞ In **theory** if the price per share given by a DCF is more than the actual share price, the company is undervalued.

And vice versa if the share price given by the DCF is less than the actual share price.

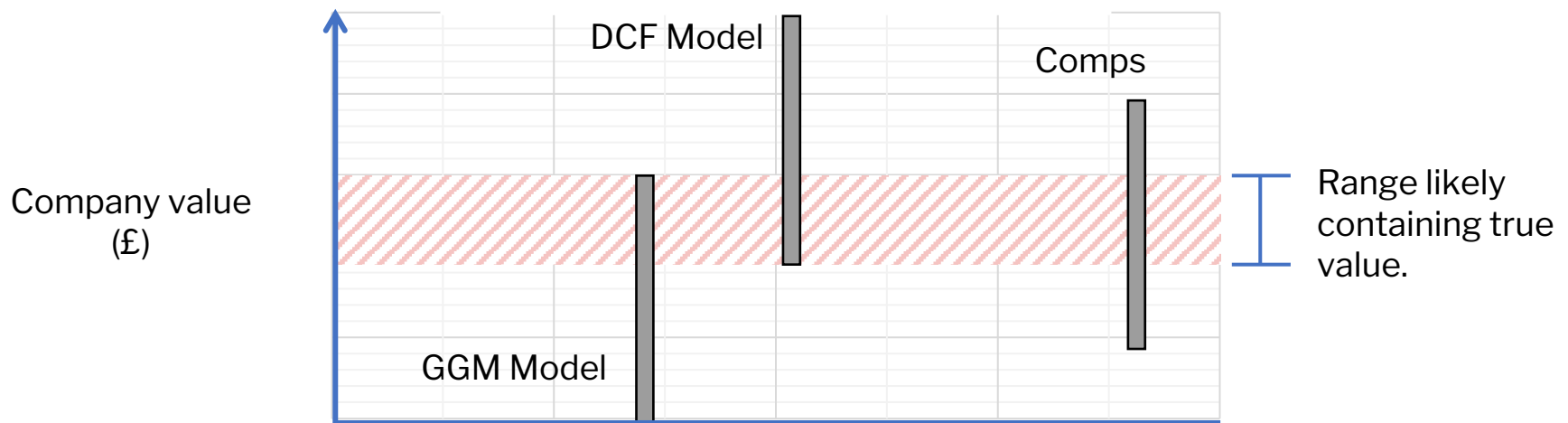
Sensitivity Analysis

We're not quite done...

☞ In the models we have discussed we have made a lot of assumptions.

☞ In order for our assumptions to not be too optimistic or pessimistic, we employ **sensitivity analysis**.

☞ This is essentially taking the worst-case and best-case scenarios for our models, and using those to provide ranges on the potential value of a company.



What's next?



👉 We will be having a live DCF model conducted by Financial Edge next week! **Tuesday, 6th February**



FINANCIAL EDGE

👉 Make sure to take advantage of discounts on their courses only available during the session!

👉 On **Thursday, 15th February** Sortino M&A group will be conducting a live LBO model!



SORTINO
M&A GROUP

👉 Make sure to bring your laptops for both sessions to follow along!



Thank you for listening!